

Best Classifier for Every Generator

Generator	Dataset	Best Classifier	Accuracy	F1	ROC-AUC
CTGAN	Cancer	Naive Bayes	0.852 ± 0.029	0.812 ± 0.034	—
CTGAN	Mushroom	Gradient Boosting	0.564 ± 0.017	0.617 ± 0.021	—
CopulaGAN	Cancer	Decision Tree	0.489 ± 0.122	0.434 ± 0.128	—
CopulaGAN	Mushroom	AdaBoost	0.566 ± 0.006	0.705 ± 0.007	—
TVAE	Cancer	Logistic Regression	0.951 ± 0.016	0.935 ± 0.021	—
TVAE	Mushroom	Extra Trees	0.625 ± 0.006	0.669 ± 0.008	—
GaussianCopula	Cancer	AdaBoost	0.910 ± 0.024	0.865 ± 0.039	—
GaussianCopula	Mushroom	Naive Bayes	0.602 ± 0.019	0.659 ± 0.017	—
WGAN_GP	Cancer	Logistic Regression	0.950 ± 0.017	0.929 ± 0.025	—
WGAN_GP	Mushroom	MLP	0.711 ± 0.011	0.757 ± 0.008	—
CTABGAN	Cancer	Logistic Regression	0.912 ± 0.019	0.874 ± 0.029	—
CTABGAN	Mushroom	Naive Bayes	0.585 ± 0.008	0.709 ± 0.010	—
TabDDPM	Cancer	SVM-RBF	0.620 ± 0.012	0.023 ± 0.057	—
TabDDPM	Mushroom	Naive Bayes	0.513 ± 0.013	0.641 ± 0.021	—
ForestDiffusion	Cancer	Logistic Regression	0.982 ± 0.011	0.975 ± 0.016	—
ForestDiffusion	Mushroom	Extra Trees	0.914 ± 0.005	0.923 ± 0.005	—